ME 333

MECHANICS OF MATERIALS

Lecture 19: Unsymmetric Bending Tutorial Problems II & Shear Stresses in Beams

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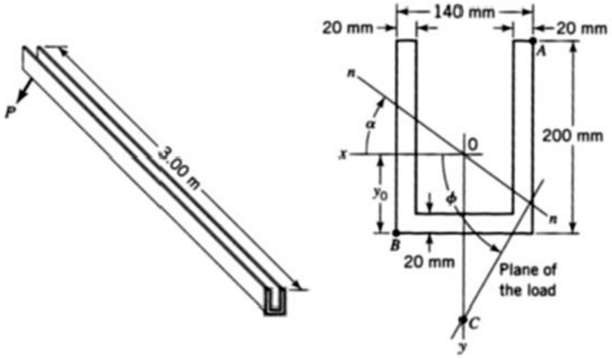
Sem I, 2025 - 2026

Unsymmetric Bending

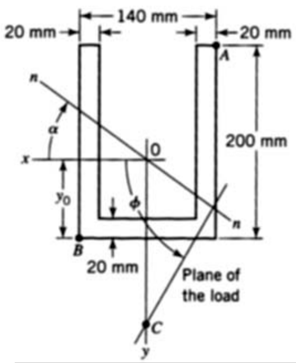
[Example 2]: Locate points of maximum tensile and compressive stresses in the beam and determine the stress magnitudes.



Ref Base Fig: Advanced Mechanics of Materials, Boresi & Schmidt

Unsymmetric Bending

[Example 2]: Step1: Orientation of N.A



- Resolve moment $M = -3P$ about x and y axis :

$$M_x = M\sin(60^\circ) = -31.18\,\text{kNm}$$
$$M_y = -M\cos(60^\circ) = 18\,\text{kNm}$$

- Determine moment of inertia:

$$I_x = 39.69 \times 10^6\,\text{mm}^4;\ I_y = 30.73 \times 10^6\,\text{mm}^4;\ I_{xy} = 0$$

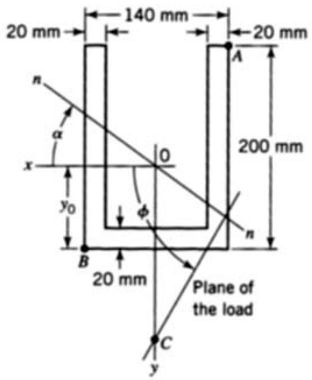
- Determine orientation of N.A.

$$-y = \left(\frac{I_x}{I_y}\cot\phi\right)x \implies \tan\alpha = -\frac{I_x}{I_y}\cot\phi$$
$$\alpha = -36.71^\circ$$

Ref Base Fig: Advanced Mechanics of Materials, Boresi & Schmidt

Unsymmetric Bending

[Example 2]: Step 2: determine the maximum stresses in the beam



- Normal stress that we derived earlier is:

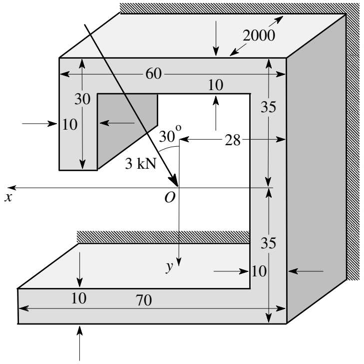
$$\sigma_x = -\frac{(yI_{yy} - zI_{yz})M_z + (yI_{yz} - zI_{zz})M_y}{(I_{zz}I_{yy} - I_{yz}^2)}$$

- Determine maximum tensile stress due to M_x & M_y :

$$\sigma_A = \frac{(yI_y)M_x + (xI_x)M_y}{(I_xI_y)} = \frac{M_x(y_A - x_A\tan\alpha)}{I_x}$$
$$= \frac{-31.18 \times 10^6(-118 - (-70)(-0.7457)}{39.69 \times 10^6} = 133.7\,\text{MPa}$$
$$\sigma_B = \frac{M_x(y_B - x_B\tan\alpha)}{I_x}$$
$$= \frac{-31.18 \times 10^6(82 - (70)(-0.7457)}{39.69 \times 10^6} = -105.4\,\text{MPa}$$

Unsymmetric Bending

[Example 3]: Find the orientation of N.A., locate points of maximum tensile and compressive stresses in the beam and determine the stress magnitudes. Given $L = 2\,\text{m}$



Unsymmetric Bending

[Example 3]: Step1: Orientation of N.A

- Resolve moment $M = 6\,\text{kNm}$ about x and y axis :

$$M_x = -M\cos(30^\circ) = -5.2 \times 10^6\ \text{N-mm}$$
$$M_y = -M\sin(30^\circ) = -3 \times 10^6\ \text{N-mm}$$

- Determine moment of inertia:

$$I_x = 1.33;\ I_y = 0.917;\ I_{xy} = 0.030 \quad (\times 10^6\ \text{mm}^4)$$

- Determine orientation of N.A.

$$-\left(\frac{M_yI_x + M_xI_{xy}}{I_xI_y - I_{xy}^2}\right)x + \left(\frac{M_xI_y + M_yI_{xy}}{I_xI_y - I_{xy}^2}\right)y = 0$$
$$\implies \frac{y}{x} = \frac{M_yI_x + M_xI_{xy}}{M_xI_y + M_yI_{xy}} = \frac{-4.15 \times 10^{12}}{-4.86 \times 10^{12}} = 0.854 \implies \alpha = \arctan(0.854) = 40.5^\circ$$

Unsymmetric Bending

[Example 3]: Step 2: determine the maximum stresses in the beam

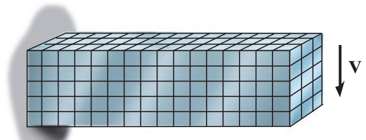
- Maximum stresses at points $A(32, -35)$ and $B(-28, 35)$ are:

$$\sigma_{zz}|_A = -\left(\frac{M_yI_x + M_xI_{xy}}{I_xI_y - I_{xy}^2}\right)x + \left(\frac{M_xI_y + M_yI_{xy}}{I_xI_y - I_{xy}^2}\right)y$$
$$= -\left(\frac{-4.15 \times 10^{12}}{1.219 \times 10^{12}}\right) \times 32 + \left(\frac{-4.86 \times 10^{12}}{1.219 \times 10^{12}}\right) \times -35 = 248.48\,\text{MPa}$$

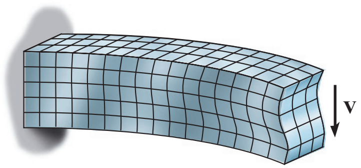
$$\sigma_{zz}|_B = -\left(\frac{M_yI_x + M_xI_{xy}}{I_xI_y - I_{xy}^2}\right)x + \left(\frac{M_xI_y + M_yI_{xy}}{I_xI_y - I_{xy}^2}\right)y$$
$$= -\left(\frac{-4.15 \times 10^{12}}{1.219 \times 10^{12}}\right) \times -28 + \left(\frac{-4.86 \times 10^{12}}{1.219 \times 10^{12}}\right) \times 35 = -234.87\,\text{MPa}$$

Stresses due to varying bending moments

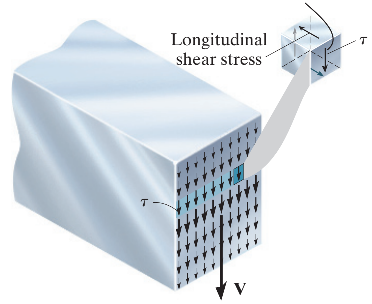
Cantilever beam with shear force acting on the free end



(a) Before deformation



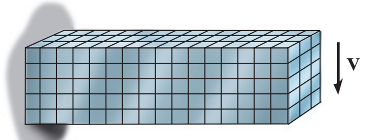
(b) After deformation



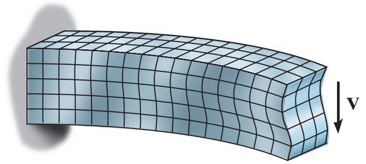
Ref Base Fig: Statics & Mechanics of Materials, R. C. Hibbeler

Stresses due to varying bending moments

Cantilever beam with shear force acting on the free end



(a) Before deformation



(b) After deformation

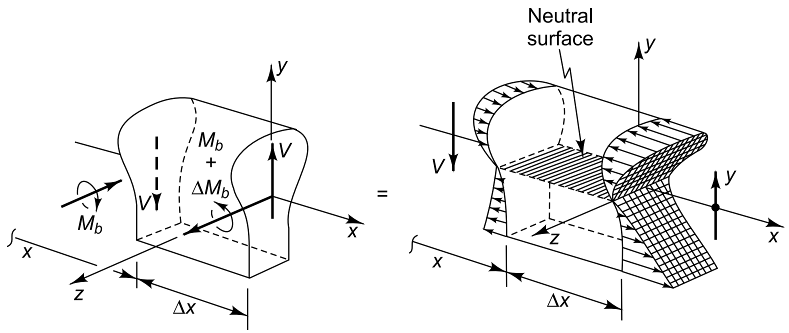
Assumptions :

- Bending stress distribution is valid even when the bending moment varies along the beam in presence of shear forces
- Satisfaction of geometric compatibility is omitted in the analysis - engineering theory
- Satisfaction of stress-strain relations is not included

Ref Base Fig: Statics & Mechanics of Materials, R. C. Hibbeler

Stresses due to varying bending moments

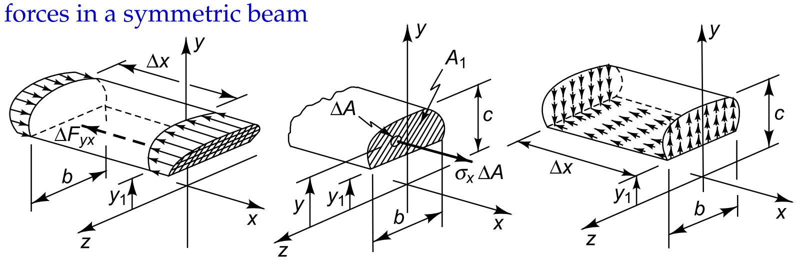
Shear forces in a symmetric beam



Ref Base Figs: An Introduction to Mechanics of Solids, Crandall et. al.

Stresses due to varying bending moments

Shear forces in a symmetric beam



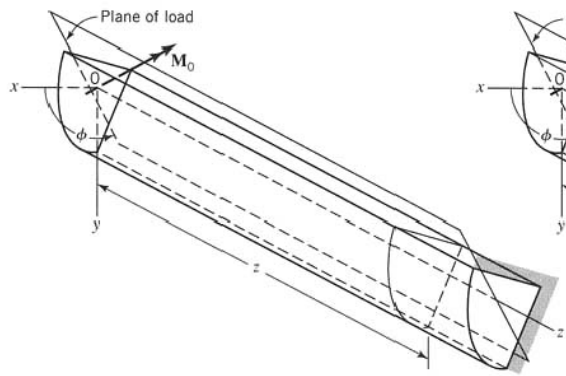
- Using static force equilibrium:

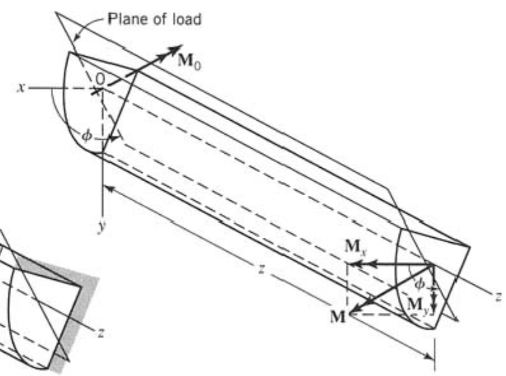
$$\frac{dF_{yx}}{dx} = \lim_{\Delta x \to 0} \frac{\Delta F_{yx}}{\Delta x} = -\frac{dM_b}{dx}\frac{1}{I_{zz}}\int_A y\,dA = \frac{V}{I_{zz}}\int_A y\,dA$$

Ref Base Figs: An Introduction to Mechanics of Solids, Crandall et. al.

Unsymmetric Bending

Pure Bending Scenario:

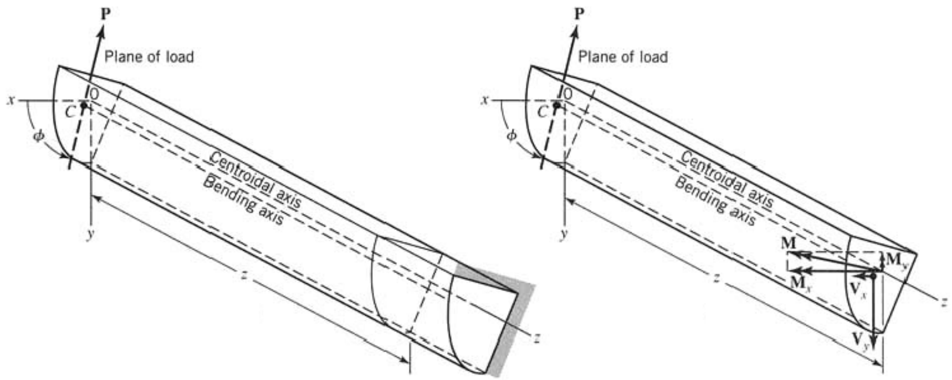




Ref Base Fig: Mechanics of Materials, Beer & Johnston

Unsymmetric Bending

Varying Bending Scenario:



Ref Base Fig: Mechanics of Materials, Beer & Johnston

Reading Assignment

Chapter 7 : Crandall et. al.
Chapters 7 & 8 : Boresi & Schmidt